It's the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic

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Abstract

Vision–language models (VLMs) are the default critic inside slide- and document-generation agents, yet they routinely pass slides whose text visibly overflows its box. We ask whether this is a failure of *perception* (the model cannot see the defect) or of *elicitation* (it sees, but the way we ask buries the signal). We settle this per defect with an attribution protocol built on a lossless structured-geometry oracle, and find the answer is not uniform: some failures are sub-perceptual under this protocol, and no prompt recovers them; others are format-suppressed and reappear under targeted defect-specific elicitation with the model and image frozen; a further reference-assisted slice recovers only when a clean twin is available. The diagnosis, not an architectural preference, prescribes the critic by routing each defect to the engine that owns its bottleneck, and a minimal symbolic–neural hybrid follows as a consequence rather than a design. On the nine image-level classes, that hybrid reaches mean balanced accuracy 0.86 against 0.59 for the conventional pointwise VLM-C0, with strict coverage reported cell by cell.

1 Introduction

An automatic critic that flags layout and content defects is now a load-bearing component of every slide- and document-generation agent, and the default critic is a general-purpose VLM asked to grade a rendered slide. These critics fail in a revealing way: shown a slide whose title visibly spills out of its box, a strong VLM under a standard rubric prompt answers "no defect." Whether this is a failure of *perception* (it cannot see the overflow) or of *elicitation* (it sees, but the way we ask buries the signal) is rarely asked, yet the answer decides whether the remedy is a larger model, a different prompt, or a different tool entirely. We treat that question as the design problem and answer it empirically.

A VLM that "cannot see" a slide defect is hiding several failures of opposite character, and separating them is the whole problem. Some defects are *sub-perceptual* (i.e., below the tested models' effective threshold under this protocol): a three-pixel alignment offset or a one-point font mismatch sits below the model's effective acuity, and no amount of prompting, scale, or resolution recovers it. Others are *format-suppressed*: the model can perceive them, but the conventional elicitation ("here is a slide and a rubric; score every defect type in one pointwise pass") buries the signal, while a targeted atomic query with required evidence recovers it with the model and image frozen. A third slice is *reference-assisted*: the model resolves the defect once a clean twin is available, but not from a single slide alone. Conflating these cases is why the field oscillates between "VLMs are bad at layout" and "VLMs can judge slides"; both hold, for different defects.

We make the split measurable with an attribution protocol built on a *lossless structured-geometry oracle*. Each slide carries an intermediate representation (IR) of element bounding boxes, text, and style; we present the same paired (defective, clean) slides under four input modalities (image-only, structured-oracle, VLM-caption, and image+oracle) crossed with detect/localize/repair tasks. Because the oracle is lossless machine-readable structure rather than a model-written caption, the perception channel it isolates is uncontaminated: a model that fails image-only but succeeds with the structured oracle is bottlenecked on perception, not reasoning. This is the methodological difference from concurrent perception/reasoning attribution that uses natural-language perception oracles, which re-inject the very perceptual error they try to factor out.

*Independent technical report. All experiments were run on a single 4×RTX 3080 (20 GB) workstation; weights, renders, and raw rollouts are not redistributed but all analysis scripts and reports are released. Correspondence: `surh6@mail2.sysu.edu.cn`.

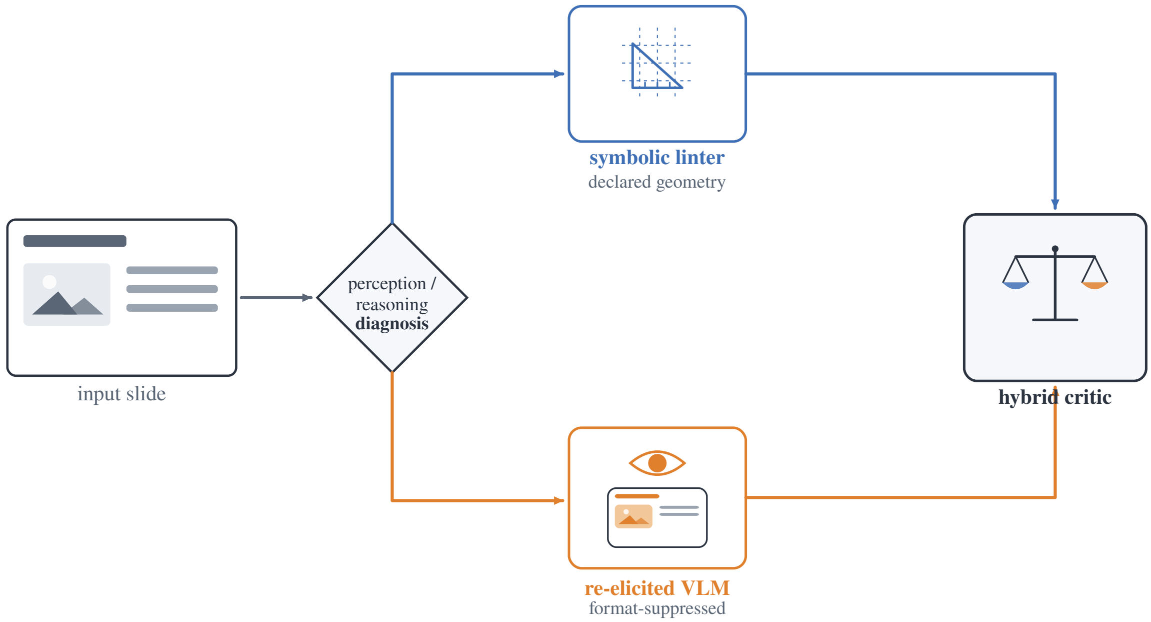

Figure 1: The paper in one picture. A perception/reasoning diagnosis (Sec. 4) shows slide-inspection failures are of two kinds. Declared-geometry defects are exact and cheap for a symbolic linter, with a sub-threshold acuity tail nothing recovers; merely format-suppressed defects are recovered by a vision–language model under a changed elicitation. Routing each defect to its bottleneck-appropriate engine yields a hybrid critic (Sec. 5). A constructed render-overflow class (legal declared box, overflowing pixels) is the crux: invisible to the linter by construction, it is caught only by engines with a rendered-quality read-out (a re-elicited VLM, a frontier VLM judge, or general perceptual reward models), not by domain-labelled scorers, so its detection tracks perceptual capability rather than training domain (Sec. 5.3).

The diagnosis, not an architectural preference, then prescribes the critic (Fig. 1): route each defect to the engine that owns its bottleneck. Sub-perceptual geometry goes to a symbolic linter that reads coordinates and rules; text and structural semantics go to a language model; and the format-suppressed, perceivable classes go to a VLM, but only under the changed elicitation the diagnosis shows recovers them. To stress-test the design we construct an adversarial render-containment overflow class that is invisible to the linter by construction (the declared box is legal; only the rendered pixels overflow) and that we find is missed by a 3B document reward and an artistic-aesthetic head as well, while two general-multimodal rewards, a perceptual-quality probe, and a closed frontier VLM-as-judge see it; its detection tracks rendered-quality read-out capability rather than the training-domain label alone. We conjecture the same declared-layout/rendered-pixel gap recurs in DOM/CSS, charts, and document layout, but do not test it here. The full symbolic–neural critic is built for *IR-owning generation agents*, where the declared structure the linter relies on is available; for bare third-party pixels the deployable critic is the re-elicited VLM alone (Sec. 7), and every coverage claim should be read within that scope.

Contributions. Our contributions form a chain. We introduce a perception/reasoning attribution protocol built on a *lossless* structured-geometry oracle (element boxes, text, and style, not a model-written caption that re-injects perceptual error) that separates a VLM's quality-inspection failures into *sub-perceptual*, *format-suppressed*, and *reference-assisted* slices, and reproduces on real CC-licensed layouts with no human or self-annotation (Secs. 4, 7). Applied across the taxonomy, the protocol yields the paper's central finding: targeted defect-specific elicitation recovers the detection that the conventional pointwise rubric buries, while G1/S6 remain a distinct reference-assisted case rather than evidence that the single-slide prompt "saw it all along" (Sec. 5). The diagnosis in turn *prescribes* the critic, rather than the reverse:

routing each defect to the engine that owns its bottleneck (a linter for declared geometry, one re-elicited VLM for the rest) raises mean balanced accuracy to 0.86 from 0.59 for the conventional pointwise VLM-C0, while the linter-blind render class makes its blind spot falsifiable and a reward-side audit reproduces the same split (Sec. 5). Supporting these, an 8B examiner finetuned on injected defects surpasses a zero-shot 30B on in-distribution semantic inspection and abstains rather than hallucinates geometry, and a perturbation-fidelity audit shows many injected geometry defects never render, a hazard any IR-labeled dataset inherits (Secs. 6, 5).

2 Related work

The VLM perception bottleneck. A growing line of work argues that VLM failures on visually grounded tasks are dominated by perception rather than reasoning, decomposing the bottleneck for chart understanding [1] or showing that the needed visual information is present in the representation but unused [2]; MMVP [3] is the canonical demonstration that VLMs stumble on minimally-different image pairs humans solve trivially, and its sub-perceptual acuity floor is what our title ("not the eyes") echoes. We share the perception-first stance but ask it of a different task family, *quality inspection of generated artifacts*, where the perception channel can be isolated exactly with structured geometry rather than approximated by captions, and where the answer feeds a critic architecture rather than a benchmark.

Perception/reasoning attribution. Concurrent work separates perception from reasoning in abstract visual-reasoning benchmarks via natural-language perception oracles [4]. We share the goal but differ in kind: our oracle is a *lossless* structured representation rather than a lossy caption, a cleaner attribution boundary because a caption re-injects the very perceptual error it is meant to factor out; our domain of generated documents admits exact injected labels; and we close the loop from diagnosis to a deployed critic. Layout-error-detection work [5] introduces distribution-grounded structural-error injection but for *document-layout-analysis predictions*, and notes that existing metrics cannot isolate whether failures stem from recognition or reasoning, the gap our attribution protocol fills.

Relative vs. absolute judgement. Our finding that relative elicitation beats absolute scoring echoes work showing VLM judges can rank but cannot reliably score [6]; that result concerns a judge's *output mode*, orthogonal to our *source of failure*, and we adopt it to motivate the pairwise and atomic-binary heads of our critic. Comparative/relative elicitation is also known to recover discriminative signal that pointwise scoring suppresses in LLM/VLM judges [7], though the protocol choice is not bias-free and can itself distort preferences [8]. Our contribution is not that effect but its *defect-class boundary*: it rescues text overflow (G1) and image/text contradiction (S6) yet not fine alignment (G3) or terminology drift (S3), which is what turns the elicitation effect into a perception/format attribution signal.

Slide generation, evaluation, and reward. Slide and poster generation with design feedback is an active area: AutoPresent [11] treats slide generation as tool-augmented program generation with iterative self-refinement, aesthetic-reward systems [9, 10] report that iterative VLM refinement helps gross layout damage but fails on fine-grained aesthetics and is prone to reward hacking, and slide-evaluation studies [12] call for calibrated critic-in-the-loop selection. These *motivate* a routed critic rather than compete with it: they observe the VLM's perceptual ceiling and route around it with rules, whereas we first *diagnose* that ceiling and show part of it is elicitation-recoverable. We probe real data with the human-annotated SlideAudit set [13] and audit five published reward/perceptual scorers spanning document, general-multimodal, and aesthetic categories plus a closed frontier VLM-as-judge (Sec. 5.3). Training specialist critics from paired data is an established recipe in natural images and code; defect injection adapts it to rendered documents at zero annotation cost, and we use it as the semantic engine of the routed critic and the perception/reasoning attribution it enables.

3 Problem setup and defect taxonomy

Task and representation. The critic inspects a single rendered slide, optionally with its declared structure, and must *flag and localize* defects of a fixed set of types; a detection counts only when it fires on a defective slide and stays silent on its matched clean twin. Every slide is an intermediate representation (IR) of typed elements with bounding boxes, text, and style; we inject defects programmatically into the IR and render, yielding exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry* group G1–G6, a *semantic* group S1–S6, and one constructed

Table 1: Defect taxonomy. The geometry group is detected symbolically; the semantic group is the examiner's domain; G7 is constructed to be linter-blind. "Channel" is the engine the diagnosis assigns (Sec. 4–5).

Group	Class	Description	Channel
Geometry	G1 text overflow	text exceeds its box	VLM (pairwise) + linter
	G2 element overlap	boxes intersect	linter
	G3 alignment offset	small mis-alignment	linter
	G4 font-size inconsistency	off-grid font size	linter
	G5 brand-color violation	off-palette color (ΔE)	linter
	G6 margin violation	element past safe margin	linter
Semantic	S1 title/body mismatch	body contradicts title	VLM
	S2 narrative-order break	deck sections out of order	LLM
	S3 terminology inconsistency	term drift across deck	term linter
	S4 density violation	slide over/under-packed	LLM
	S5 missing logic section	dropped argument step	LLM
	S6 image/text contradiction	figure contradicts caption	VLM
Render	G7 containment overflow	legal box, rendered overflow	VLM (atomic-binary)

render class G7; it is aligned to the expert-derived SlideAudit taxonomy [13] via a bidirectional map, and results are reported in its canonical classes (Appendix A). We add G7 because it is the sharpest test of the linter's blind spot: the declared box is legal (inside the safe margin, no overlap) but the rendered content overflows its container, so the defect exists *only* in pixels. On real agent-generated decks G7 is common, affecting most slides of a Zenodo10K slice and over half of an AutoPresent slice.[1]

Attribution modalities and metric. We present each paired sample under four modalities: **A** image-only; **B** the lossless structured oracle; **B′** a fixed VLM-written caption; and **C** image+oracle, each crossed with detect/localize/repair (Fig. 2).[2] The attribution logic is operational: A-fail ∧ B-succeed means the class becomes solvable once the same information is made explicit as lossless structure, so we route it as *perception-bottlenecked*; A-fail ∧ B-fail means the class is unsolved even at the structured interface and is routed symbolically. The headline metric throughout is *balanced accuracy on paired clean controls*, never recall alone, which a model can saturate by always flagging.

Two linters. A geometry linter detects G1–G6 from coordinates and rules (brand color via CIELAB ΔE_{2000}, not RGB distance), at zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus near-duplicate clustering, image-free. Both are the "detectors of record" the diagnosis routes to.

4 Diagnosis: two kinds of blindness

Pointwise image-only inspection floors fine geometry. Under pointwise image-only inspection, G2–G6 sit at chance for 4B and 8B models, and a 30B model breaks only the coarsest class (G1 overflow). This pointwise floor is invariant to the things one would tune first: swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a native-resolution NaViT, and MoonViT) leaves every model at the random line, differing only in abstain-vs-over-flag bias, and raising input resolution from 1536 to 2048 changes nothing. The lever is the elicitation and the defect magnitude, not the encoder or the pixel budget (the next paragraph and Fig. 5 show targeted/relative elicitation lifts supra-threshold alignment well above chance), so the floor is a property of the *pointwise* protocol, not a hard capability limit. Either way the right detector for declared geometry is the symbolic linter: it reads the coordinates directly at balanced accuracy 0.75–1.0 with zero false positives, cheaper and exact where the IR is available, not because the VLM cannot see the defect.

[1] 93-deck Zenodo10K slice: 81/93 decks affected (857/3167 legal boxes); 14-deck AutoPresent slice: 8/14 decks (20/50 legal boxes).

[2] The oracle's ground-truth markers are stripped by a single regression-tested de-leaking view. B′ is a lossy-caption control, not a second oracle: because it is VLM-produced, B's advantage for isolating declared structure is a property of the channel by design, not an empirical claim that captions are always worse.

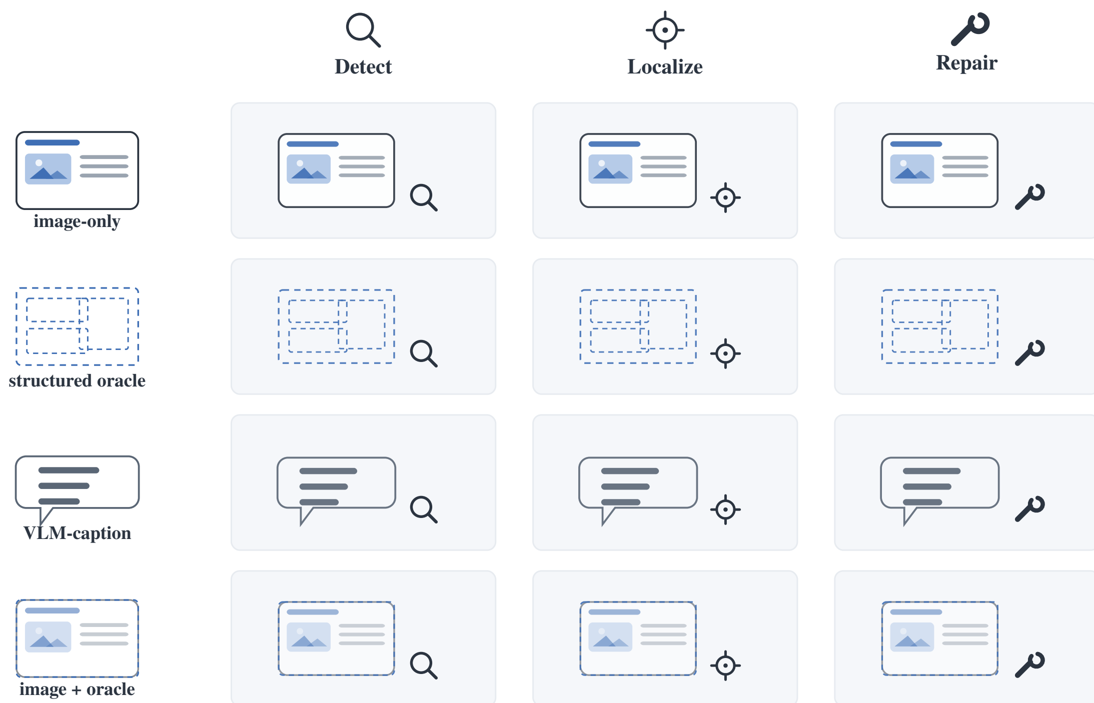

Figure 2: Attribution protocol. Each paired (defective, clean) slide is presented under four input modalities (image-only, the lossless structured oracle, a fixed VLM-caption control, and image+oracle) crossed with detect/localize/repair. Comparing image-only against the structured oracle is what isolates a clean perception/reasoning split (Sec. 4).

But some apparent perceptual failures are elicitation artifacts. The same pointwise image-only setting hides a second, opposite phenomenon. For G1 text overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50, indistinguishable from the genuinely sub-threshold tail, yet a two-alternative forced choice between the defective and clean slide jumps to 1.00 (Fig. 4): the defect is perceivable, but the single-slide pointwise call does not surface it. We therefore treat G1/S6 as *reference-assisted*: recovery needs the clean twin, not just a renamed single-image question. G3 alignment behaves differently, but only *above a magnitude threshold*: sweeping the injected offset as an internal contrast (one element shifted from its aligned sibling column, decidable from the slide alone), the forced choice saturates to 1.00 by 16 px (0.83% of slide width) while the sub-threshold tail ($\leq$ 8 px) stays at chance even side by side (Fig. 5). So fine alignment is not a class-level capability floor: above threshold it is format-suppressed and recoverable, with a sub-perceptual residue only in the small-offset tail. (*How* the elicitation helps differs by class, a named target versus a paired clean reference, which we decompose in Sec. 5.1.) The connecting thread is a *magnitude-gated* recoverable-versus-sub-perceptual boundary rather than a clean geometric/semantic split, and S3 terminology (below) remains the one defect no elicitation rescues.

On the rebuilt corpus we also now have a human spot-check on the actual pairs cited here, not just the earlier diagnosis runs. Replacing only the stale legacy G3/G5 samples and keeping the other 55 pairs fixed, the v2 pass verifies every sampled defect as visible and every clean twin as clean (73/73, Wilson lower bound 0.95 for both), with source-structure faithfulness 55/55 on the auditable classes. The delta is exactly where the re-operationalisation says it should be: G3 shifts from 0/7 on the old external-reference sample to 8/8 on the corrected within-list misalignment sample, and G5 from 0/7 to 10/10 on the corrected within-list chromatic clash sample; the non-replaced 55 pairs are unchanged.

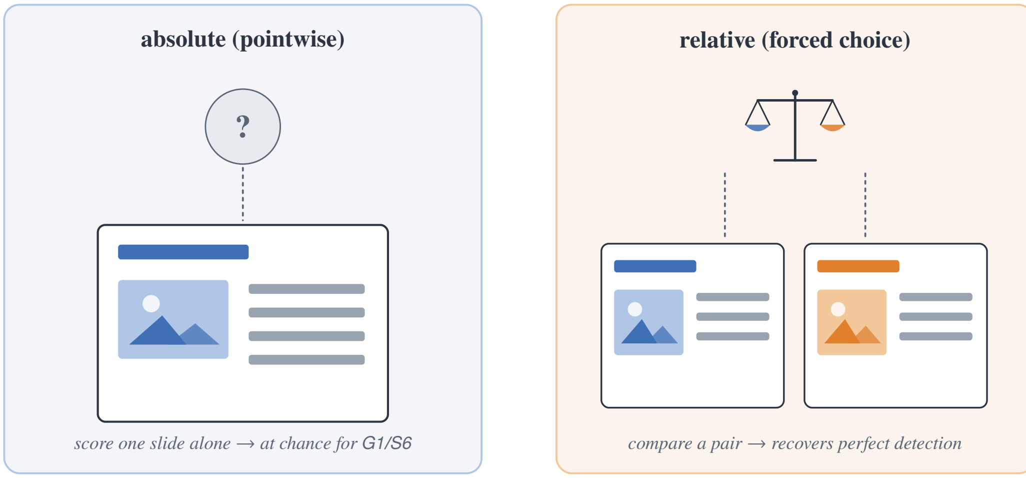

Figure 3: Two elicitations, same model and images. Pointwise (absolute) scoring inspects one slide in isolation; the forced choice compares a pair. For G1 and S6, changing absolute to relative is what recovers detection; quantified in Fig. 4.

A counterexample: terminology. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. The structured-oracle channel, which feeds the model the full deck text with both variants present, also fails (0.56): with both variants already explicit in the input, this failure cannot be OCR, and instead reflects cross-page term-matching that does not survive the yes/no framing. The image channel is consistent with fine-grained glyph discrimination and the oracle channel fails even with the text in hand, so neither route is reliable. S3 leaves the VLM entirely for a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM's 0.69, a negative result that *re-routes* the class.

A genuine geometric blind spot: G6 page offset. The re-operationalisation does not rescue every class; on one it fails outright. We re-pose G6 margin as a *page offset*: the whole content block is shifted toward one edge (every element together, so internal alignment is preserved and the defect cannot be misalignment), leaving one margin crowded against (or running off) the edge and the opposite side empty, decidable from the slide alone.[3] Across all six VLMs and both pointwise framings (absolute "past the safe margin" and relative "shifted to one side") the page offset stays at chance (roster-mean C3 0.50), and the discriminative 2-AFC, the very test that separated *data artifacts* (S6 image/text contradiction, which recovers to perfect once the figure is actually rendered) from genuine blindness, leaves G6 on the floor for *every* model (Fig. 6): none can tell a page-offset slide from a balanced one until the content is clipped by the page boundary. So unlike supra-threshold G3/G5 (format-suppressed) and G1/S6 (reference-assisted), page offset is a *genuine* perceptual blind spot (VLMs do not represent global content positioning), which is why declared geometry routes to the linter.

A methodological aside that becomes load-bearing. Enterprise "snap-to-master" templates re-fit element boxes to a grid before rendering, and we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping, while overflow, driven by content-determined text length, survives. This collapses the inspectable spectrum toward two poles, pure-perception defects templates cannot constrain and pure-semantic defects they do not touch, and is a trap for any dataset that derives labels from the IR but evaluates on the rendered image, a hazard we quantify and tie to a reward-model failure in Sec. 5.3.

[3]The original S6 render omitted the figure element, so the cross-modal contradiction was literally unseeable (recall 0); on the valid-figure corpus the S6 2-AFC reproduces at 0.75–1.00 across all six VLMs, confirming it is a render artifact, not blindness.

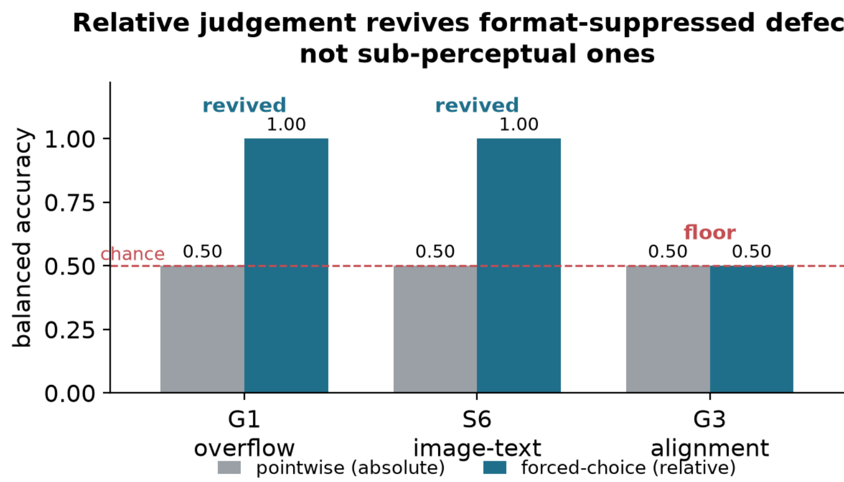

Figure 4: Relative judgement recovers absolute-pointwise blind spots. For G1 overflow and S6 image/text contradiction, pointwise (absolute) inspection is at chance but a forced choice recovers perfect detection (*elicitation-recoverable*). G3 alignment behaves the same way above a magnitude threshold (forced choice 1.00 once the offset clears $\approx 0.8\%$ of slide width; Fig. 5); only its sub-threshold tail and S3 terminology stay at chance side by side. Same model, same images; only the elicitation changes. What *drives* the recovery (naming vs. a paired reference) is decomposed in Fig. 8.

5 From diagnosis to a symbolic–neural critic

The diagnosis says a single model is the wrong unit: route each defect to the engine that owns its bottleneck. We build that critic, validate the elicitation recovery it relies on across many models, and stress it with the constructed render class G7.

5.1 Elicitation: format suppression is not capability

Four ways to ask. We compare four elicitations holding the model and image fixed: **C0**, the conventional pointwise+rubric pass over the whole taxonomy in one call; **C1**, free-form describe-then-classify; **C2**, a geometry-normalized synthetic-twin pairwise (re-render snapped-to-grid, both orders); and **C3**, an atomic per-type yes/no with forced localization. The scientific contrast is C3 vs. C0: same model, taxonomy, and image, differing only in "ask everything at once" versus "one atomic binary with forced evidence." If C3 $\gg$ C0, the pointwise abstention was a format/cognitive-load artifact, not a capability gap.

C3 recovers the linter-blind class across the capable subset. We scope the claim to models that can see G7: a model is in-scope only if C3 clears the paired-clean detection gate (BAL-ACC ≥ 0.70, precision ≥ 0.70) and localizes the overflowing element, which excludes InternVL-8B and Ovis-9B as capability-boundary cases. On the remaining four instances across three scales and three vendors, atomic-binary C3 recovers detection that C0 suppresses (Fig. 7: 0.50–0.79 $\rightarrow$ 0.93–1.00, precision 0.88–1.00). The effect we claim is *cross-model*: a stratified (Cochran) McNemar over the four models' matched pairs counts 233 G7 items C3 fixes against 2 reversals (χ^2_1=227, $p < 10^{-50}$).[4] The gate is defined by C3, but this cross-vendor test is gate-independent, so the recovery is not a definitional artifact. The mechanism: C0's fixed whole-taxonomy rubric cannot *name* an off-taxonomy render class, so the model abstains; the atomic per-type binary restores it.

It is the format, not the eyes. Does naming G7 suffice, or does the *atomic format* do the work? We separate them with a **C0+** control (the C0 whole-taxonomy call with G7 *added* to the catalog, otherwise byte-identical) and a single-slide *C0_named* condition that names the class on an isolated atomic prompt. The recovery factors into coverage and format. *Coverage*: naming recovers *recall*, since C0→C0+ flips 234 defectives from un-named to named against 0 reversals

[4]Per-model exact McNemar $p \leq 1.5\times10^{-11}$, all Holm-surviving over the pre-registered 85-test family; the two reversals are on Gemma4-31B (C3 recall 88/90). Full corrected table in the released multiplicity report.

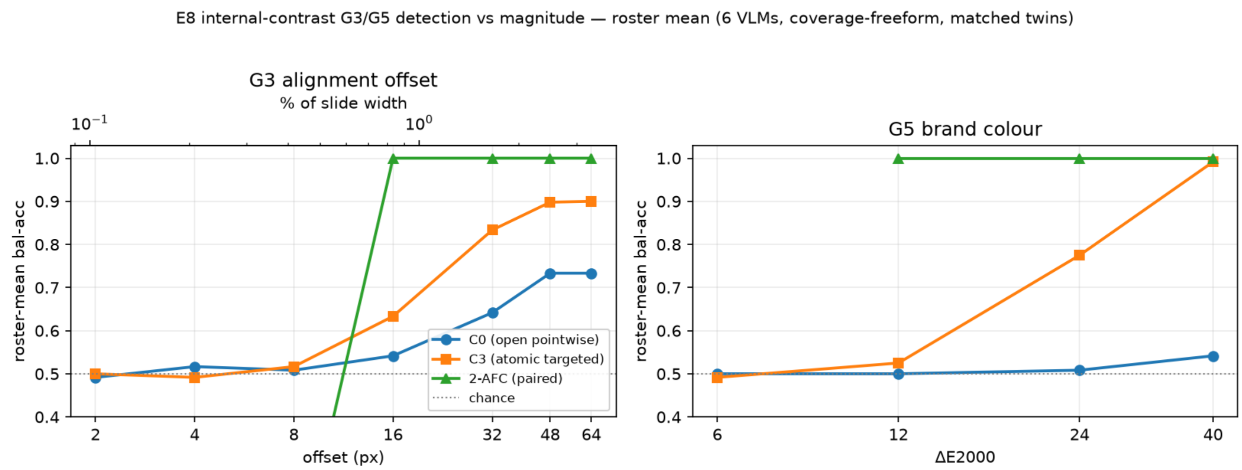

Figure 5: Fine geometry is magnitude-gated, not a flat capability floor. Roster-mean balanced accuracy (6 VLMs, matched-twin internal-contrast set) versus injected magnitude for G3 alignment (left; px and % of slide width) and G5 brand color (right; ΔE_{2000}). Open pointwise (C0) lags; atomic targeted elicitation (C3) recovers to a ≈ 0.90 plateau by 48–64 px / 0.99 by ΔE 40; the paired 2-AFC saturates to 1.00 by 16 px (0.83%) / ΔE 12. The sub-threshold tail (≤ 8 px, $\leq 6\,\Delta E$) stays at chance under every protocol, the genuinely sub-perceptual residue. Single-slide C3 plateaus below 1.0, a residual ceiling on *absolute* spatial judgement that contrastive elicitation does not share.

($p{=}7{\times}10^{-71}$; named-recall 0.00 → 0.64–0.97). *Format*: but naming *inside the overloaded rubric* wrecks specificity, since C0+ over-flags clean twins on 55% of pairs (183/330) versus 2.4% (8/330) for C3, so C0+ balanced accuracy stalls at 0.58–0.82 while C3 reaches 0.93–1.00. Naming the same class on the *atomic* prompt (C0_named) instead recovers full balanced accuracy, and a paired clean reference adds nothing on top (paired-clean Holm $p \leq 0.002$): so for G7 the suppressor is the whole-taxonomy *format*, not a lack of visual signal. Forced evidence confirms real perception rather than a lucky "yes": on the capable models' G7 true positives the named region and overflowing element are correct ≥98%, and the models read back the specific spilling content ("the list items starting from 'Cost attribution' "), which a bluffing model cannot do. The same decomposition draws the boundary: for declared geometry G1 and image/text contradiction S6, naming alone does *not* help (C0_named ≈ 0.50) and the recovery needs the paired reference (2-AFC → 0.97–1.00), a milder *reference-assisted* effect, not format suppression. A clean-vs-clean guess-floor rules out a forced-pick artifact for both: capable models answer "tie" on 92–100% of clean pairs (Fig. 8).

Compute-matched control: recovery without additional test-time compute. The remaining alternative is that C3 wins only because it spends more test-time compute than the one-call rubric. We test that directly on three API-served models spanning distinct vendors, holding the image and rendered corpus fixed and matching C3's budget with repeated C0 draws. The headline control is *self-consistency*: majority vote over $K{=}10$ independent C0 samples. On G7, C3 still beats that compute-matched control on all three vendors (0.92/0.88/0.83 vs. 0.61/0.59/0.63; $\Delta{=}+0.31/+0.29/+0.20$), and all three contrasts survive Holm correction within the full 24-test family (reported in the supplement). The budget points the same way: C3 is the cheapest condition in completion tokens (34.7/37.5/40.5 output tok per slide, one call) while the matched self-consistency baseline spends 8–10 calls per slide and still loses. The result is therefore not that a heavier budget rescues the rubric; it is that the whole-taxonomy pointwise format suppresses detection, and breaking that format recovers it without additional test-time compute. We report the union aggregation of the same draws, the definition-matched one-call control, and the G1 negative control in the supplement.

No universal elicitation, so the critic routes. The best elicitation is jointly set by model capability and defect class. C2 synthetic-twin pairwise (not C3) rescues *declared* geometry G1 on capable models (Qwen3.5-27B 0.69 → 0.93, Qwen3.6-27B 0.53 → 0.78, precision 1.00), because snapping absorbs the declared overflow and the contrast exposes it; and C1 free-form, which rescues weak abstaining models, *collapses* on strong ones, which free-associate nits and flag nearly every clean slide (precision ≈ 0.50). There is no one prompt, which is why the critic routes per defect and per model. The format effect is not a synthetic artifact: on the human-annotated SlideAudit set, with no structure available, C3 still beats C0 for every class (brand color 0.52 → 0.81, alignment 0.56 → 0.72). Absolute geometry on bare real

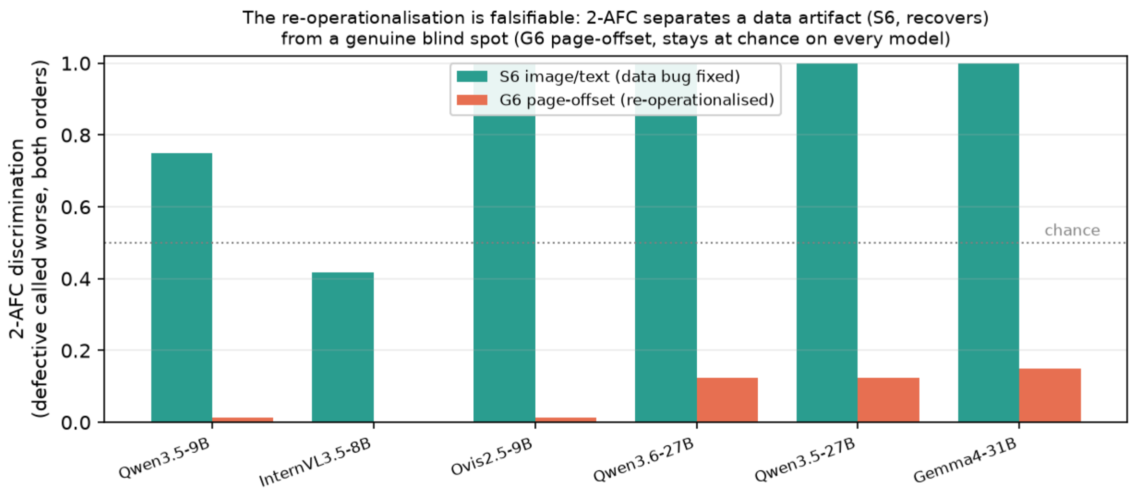

Figure 6: A genuine blind spot the forced choice exposes. Per-model 2-AFC discrimination (fraction of paired slides where the defective is consistently called worse in both presentation orders; ties count as not-discriminated). Single-image pointwise puts both classes at chance, but the forced choice *separates* them: the data artifact S6 (image/text contradiction, on the fixed valid-figure corpus) recovers on every model, while the re-operationalised G6 page offset stays at the floor on every model: a genuine blind spot, not an elicitation or data artifact.

pixels stays modest (0.55–0.74 outside blatant overlap): real fine geometry needs the linter, hence the element structure bare images lack, so on image-only real data the hybrid runs in a degraded VLM-only mode, a ceiling set by missing structure, not missing method.

5.2 The minimal hybrid critic covers the most

Complementary coverage the diagnosis predicts. We evaluate five critics by *named attribution*: the correct defect must be emitted on the defective slide and not on its clean twin, and a class is "covered" when named-attribution balanced accuracy exceeds 0.75 (Wilson lower bound > 0.65). Coverage follows what the diagnosis prescribes (declared geometry to the linter, the format-suppressed classes to a C3-elicited VLM), and the headline is their *combination*: the linter alone covers 4/9 classes (mean bal-acc 0.66) and VLM-C3 alone 3/9 (0.72), but *linter* $\oplus$ *C3-VLM* covers 6/9 at 0.86, matching the pre-registered routed hybrid (6/9, 0.85), against the conventional VLM-C0's 1/9 (0.59) (Fig. 9, Table 2). The nine classes are the image-level paired-clean classes; G4/S3 are linter-owned and S2/S5 are deck-scope (Sec. 6). A control settles why this is the headline: per-class routed prompt/LLM tuning is *not* load-bearing, since the routing rule follows from the diagnosis, not from optimizing over class outcomes, so the coverage is what the diagnosis predicts, not a tuned architecture.

A data-driven routing correction. We initially routed S1 title/body mismatch to the text LLM, but the text-only probe over-flags (0.25, precision 0.09) while the image-bearing VLM names it reliably (0.94, precision 0.90): title/body mismatch needs the *rendered layout*, not just the text. We re-route S1 to the VLM under its *open* pointwise prompt (C0, 0.94) rather than the atomic C3 (0.83): recall is already saturated (both catch all 18 defectives), so C3's forced yes/no only adds false positives, dropping specificity 0.89 → 0.67, a concrete case of the per-class elicitation routing of Sec. 5.1. The one image-level class the deployable hybrid does not cover is S6 image/text contradiction, at 0.50 single-image for every engine; the 2-AFC forced choice (Fig. 6) shows the model *does* perceive it (recall 1.0, specificity $\sim$ 0, over-asserting rather than missing), so this is a *reference-assisted* effect: it needs the paired clean twin (Fig. 8), not a blind spot, in pointed contrast to the genuine G6 page offset.

5.3 G7: the linter-blind class that specialised rewards also miss

Construction and headline split. G7 makes the linter's blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text; the overflow

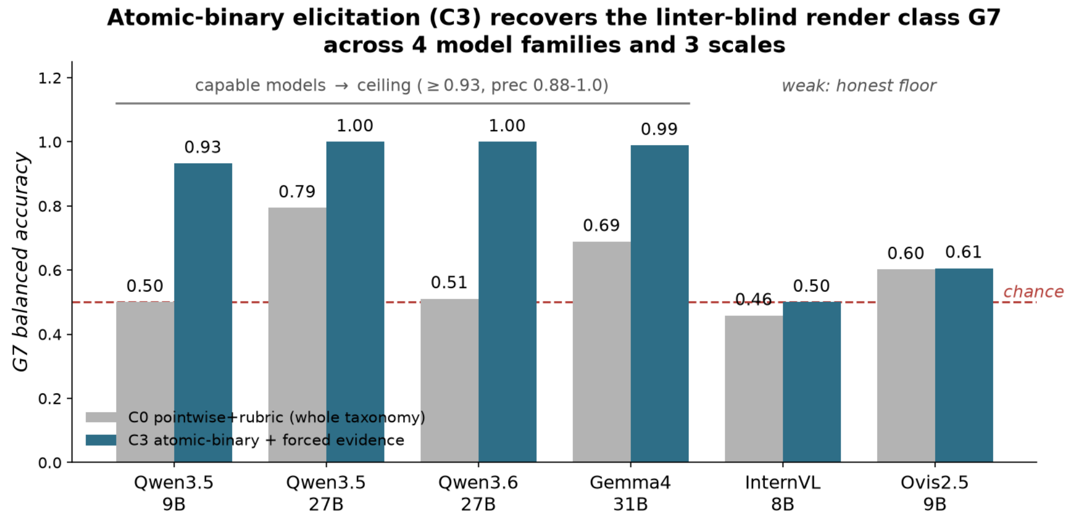

Figure 7: Atomic-binary elicitation (C3) recovers the linter-blind render class G7 across the capable subset. Capable models reach the ceiling (≥0.93, precision 0.88–1.0); the weaker InternVL/Ovis stay near their perceptual floor, a capability-dependent boundary, not a universal fix.

lives in the renderer, so the linter and any structure-only model are blind by construction—a self-check confirms zero findings on 90/90 defectives. Auditing five published scorers plus one frontier VLM judge on the same paired slides gives a decisive same-backbone dissociation: CLIP-IQA detects G7 (0.83, $[0.74, 0.90]$), while the LAION aesthetic head on the same CLIP-L/14 is at chance (0.57, $[0.46, 0.66]$). Across the scorers tested, G7 therefore tracks a rendered-quality read-out rather than the training-domain label. The complete per-scorer table and the instance-level DocReward caveat are retained in the submission supplement.

Technical-report audit detail. The natural rebuttal is "just train a neural reward model" or "use the frontier VLM as the judge." We test both directly by auditing five published reward/perceptual scorers plus one closed frontier VLM-as-judge on the same paired slides: does each score the clean slide above its defective twin? The scorers span three categories. *Document reward*: DocReward [14] (Qwen2.5-VL-3B, Bradley–Terry head over 117K paired documents). *General-multimodal rewards on distinct backbones*: Skywork-VL-Reward-7B [15] (Qwen2.5-VL-7B value head) and PickScore [17] (CLIP-H/14 contrastive reward, 500K human choices). *Aesthetic/perceptual scorers*: LAION-Aesthetic [16] (linear artistic-appeal head on CLIP-L/14) and CLIP-IQA [18] (zero-shot "Good/Bad photo" quality probe on the same CLIP). The judge is Qwen3.7-Max on DashScope (pointwise 0–100 visual/layout-quality score). Each penalizes several in-taxonomy defects it can see (Table 3), confirming each is a functioning scorer, not a domain mismatch.

The G7 split: perceptual capability, not the domain label. On G7 the picture splits, but *not* along the domain label. The cleanest dissociation is within one backbone: CLIP-IQA detects G7 (0.83, $[0.74, 0.90]$, n=90), while the LAION aesthetic head on the same CLIP-L/14 is at chance (0.57, $[0.46, 0.66]$). The same visible overflow is detected by *both* general-multimodal rewards on distinct backbones (Skywork-VL 0.79, $[0.69, 0.86]$, Holm $p=2\times10^{-6}$; PickScore 0.71, $[0.61, 0.80]$, $p=2.5\times10^{-3}$) and by Qwen3.7-Max as a pointwise VLM judge (1.00, $[0.86, 1.00]$, n=24); it is missed (CI spans 0.5) by the symbolic linter (0.00), the artistic-aesthetic LAION head (0.57), and the document reward (DocReward 0.48, $[0.38, 0.58]$). The full split is in Table 3. (Naming containment in Skywork's prompt lifts G7 to 0.87, reinforcement rather than rescue. On G5 (re-operationalised as an internal chromatic clash, visible from the slide alone) *most* scorers react (0.65–0.78), as the perceptual-capability reading predicts; the linter still owns G5 at 1.00 by exactness and palette knowledge, not because the clash is imperceptible.)

What the split implies. The DocReward miss is *instance-level*, not categorical: with a single 3B model capacity and training objective are confounded, and a larger or differently-trained document reward may well detect G7. The supported claim is narrower and stronger: G7 is visible to general rendered-quality read-outs but can be *suppressed*

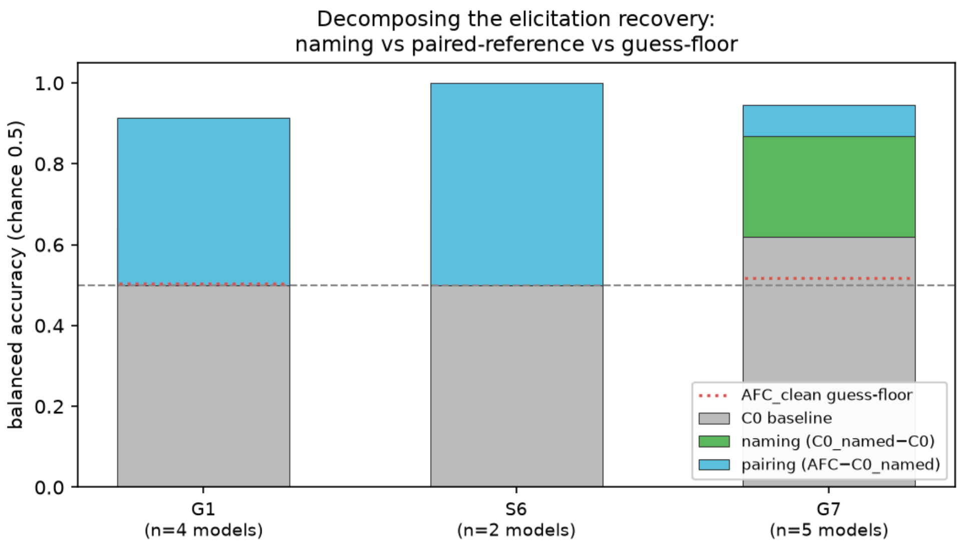

Figure 8: What drives the recovery: naming vs. a paired reference. Holding model and image fixed on the freeform set, the chance→ceiling jump factors into a *naming* component (C0_named−C0: one slide, the defect named, *no* reference) and a *pairing* component (AFC−C0_named: a clean twin supplied). For the constructed render class G7 the naming term carries it (green); for declared geometry G1 and semantic S6 the paired reference does (blue). The AFC_clean guess-floor (dotted) is near chance, so neither is a forced-pick artifact. Mean over the models that recover (AFC ≥ 0.7).

by a specialised read-out on the *same* backbone, and it is the reward-side counterpart of the C0→C3 recovery: the perception is present; the read-out, like the prompt format, can bury it. The hybrid thesis is unchanged: G7 needs a perceptually-capable engine, whether prompted, judged, or reward-headed. Elicitation stays load-bearing even for the frontier judge: its whole-taxonomy prompt cannot *name* the off-taxonomy class (C0 0.50) while C3 names and localizes it perfectly (1.00).

Perturbation fidelity: 45% of injected geometry never renders. The reward audit ties to the template result of Sec. 4. Rendering every injected defective two ways (faithfully and snapped-to-master), we find 45% of injected geometry defects (100% of overlap, alignment, margin; 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to *all five* reward scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: the two images are pixel-identical, so no critic (document, general-multimodal, or aesthetic) can react, a dataset-integrity hazard (identical inputs force identical scores), not a reward-model deficiency. This part *is* model-agnostic by construction, and any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not assumed from the IR, a prerequisite the field largely skips, and the reason our elicitation experiments render defectives faithfully.

6 The semantic engine: a specialist examiner

The hybrid's semantic engine can be a zero-shot model, but a small specialist does better on its own ground. We finetune Qwen3-VL-8B with QLoRA on 2,142 injected supervision records; the submission supplement summarizes the training mix and full evaluation table.

An 8B examiner surpasses a zero-shot 30B on *in-distribution* semantics. On in-domain held-out page-semantic inspection, the finetuned 8B reaches balanced accuracy 1.000 on both S1 title/body and S4 density, beating both zero-shot baselines including the 30B (0.93 / 0.77). The largest gap is S4 density (1.00 vs. 0.77 for the 30B), and both in-distribution contrasts survive Holm correction over the 85-test family. Thus, on the examiner's native in-distribution semantic tasks, the specialist 8B beats the larger zero-shot model; full format-control and severity results are in the supplement.

Table 2: Per-class named-attribution balanced accuracy with 95% Wilson intervals; n = paired positives with matched clean controls (precision per cell in the released reports). The minimal linter+VLM-C3 hybrid matches the routed coverage and slightly exceeds its mean. The small S1 (n=18) and S6 (n=12) cells are routing diagnostics, not confirmatory tests: the S6 reference-assisted and S1 image-routing claims rest on the elicitation decomposition (Fig. 8) and the 2-AFC discrimination, not on these cells.

Class	routed to	linter	VLM-C0	VLM-C3	linter+C3	routed	n
G1 overflow	linter	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	40
G2 overlap	linter	0.80 [.68,.87]	0.71 [.60,.79]	**0.86 [.74,.92]**	0.80 [.68,.87]	0.80 [.68,.87]	40
G3 alignment	linter	0.93 [.85,.96]	0.62 [.55,.68]	0.71 [.63,.77]	**0.93 [.85,.96]**	**0.93 [.85,.96]**	70
G5 color	linter	1.00 [.91,1]	0.50 [.46,.54]	0.68 [.57,.75]	**1.00 [.91,1]**	**1.00 [.91,1]**	40
G6 margin	linter	0.75 [.67,.80]	0.51 [.40,.61]	0.51 [.47,.55]	**0.75 [.67,.80]**	**0.75 [.67,.80]**	80
G7 render-overflow	VLM (C3)	0.00 [0,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	**1.00 [.91,1]**	40
S1 title/body	VLM (C0)	0.50 [.41,.59]	**0.94 [.75,.98]**	0.83 [.63,.92]	0.83 [.63,.92]	**0.94 [.75,.98]**	18
S4 density	LLM	0.50 [.45,.55]	0.51 [.44,.59]	**0.93 [.81,.97]**	**0.93 [.81,.97]**	0.72 [.60,.80]	36
S6 image/text	VLM	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	12
mean / classes covered		0.66 / 4/9	0.59 / 1/9	0.72 / 3/9	**0.86 / 6/9**	0.85 / 6/9	

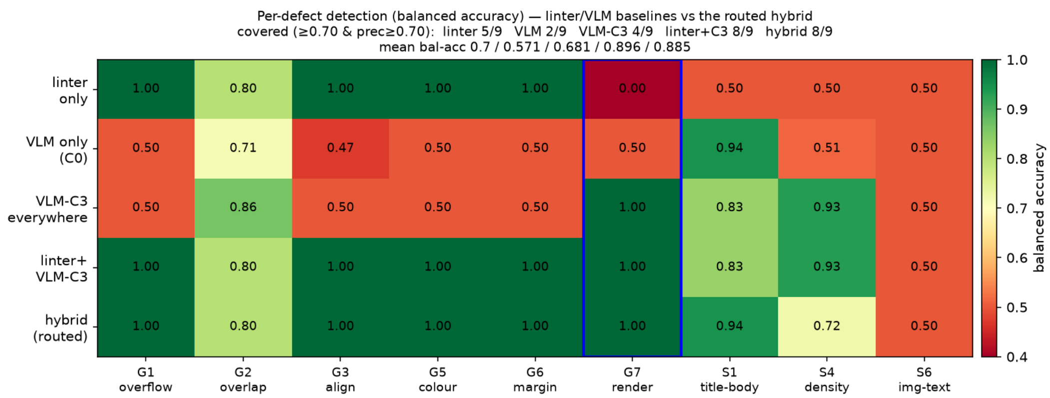

Figure 9: Coverage by engine. Per-defect balanced accuracy for the linter, VLM-C0, VLM-C3, linter+VLM-C3, and the routed hybrid. Minimal linter⊕C3 matches routed coverage; G7 (boxed) is the linter-blind cell rescued by C3.

It abstains instead of hallucinating geometry. Under image-only inspection, the finetuned model stays at 0.50 on G2–G6 with *zero* false positives: trained with image-only abstention targets, it declines to hallucinate geometry from pixels and leaves those classes to the linter, the designed behavior, and the reason the hybrid's geometry column comes from the linter rather than the VLM.

The main negative result is sim-to-real. The examiner does not transfer as a bare-pixel real-world inspector. On image-only SlideAudit, its synthetic S4 strength collapses to chance (0.50), whereas only the zero-shot 30B shows real density signal (p=1.1×10^{-5}). Scale, not finetuning, carries bare-pixel transfer: the examiner learned a distribution, not universal slide quality.

7 External validity

A real pre-sales agent deck. We run the same five-level examiner gradient on a real 16-slide proposal produced by a deployed PowerPoint agent. Its verifiable defect is 20 unfilled template placeholders ("add a title here") across 7 of 16 slides, a semantic-completeness defect the geometry linter is blind to by construction (the boxes are geometrically fine). The result reinforces the thesis from an unexpected angle: the zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the *finetuned* 8B, top of the intrinsic-quality ranking, is among the *worst* on this out-of-distribution defect (names it 0/7), and the linter's apparent recall is spurious

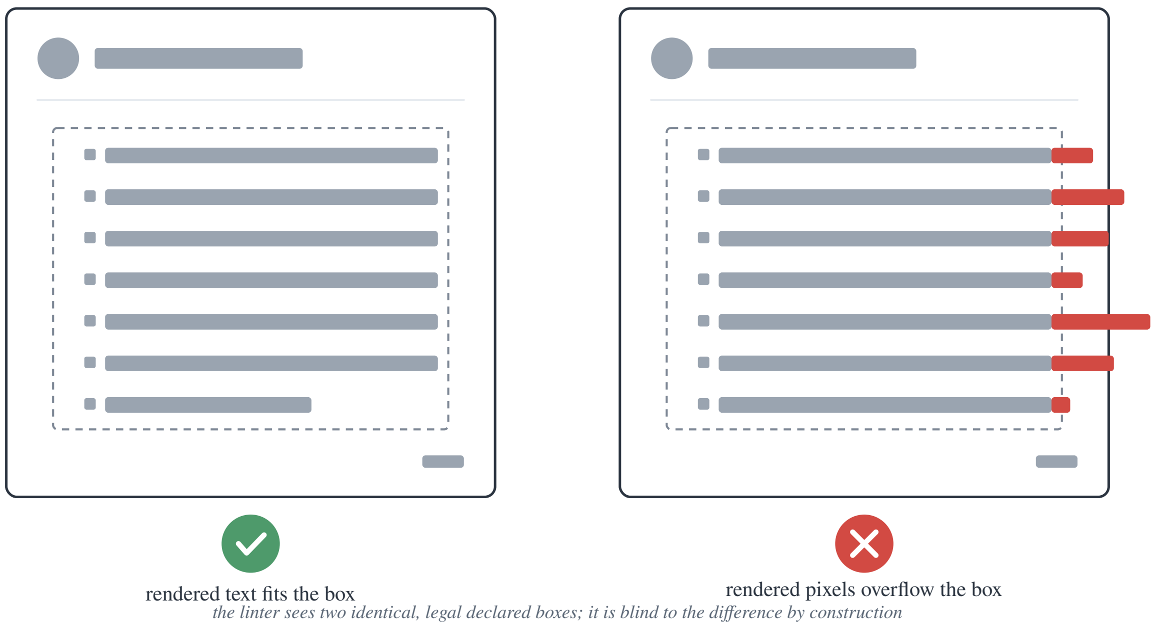

Figure 10: G7 render-overflow. The declared box (dashed) is legal and *identical* in both slides; only the rendered pixels overflow on the right. The geometry linter and any structure-only model read the same legal box in both, so they are blind to G7 by construction; only an engine with a rendered-quality read-out catches it.

geometry noise with zero placeholder mentions. The ranking on a novel defect is by general-VLM ability, not by the specialist's in-distribution quality: the examiner learned a distribution, not universal quality. Feeding the flagged placeholders back to the agent's own generator for one revision pass drives the verifiable defect from 20 to 0.

The structured attribution holds on real layouts. The body runs the A/B/C attribution on synthetic slides; the cleanest discrimination needs it on *real* geometry with a *real* oracle, which seemed to require human or model annotation. We obtain it without either: Zenodo10K [19] ships real CC-licensed `.pptx` with intact element XML, so `python-pptx` extracts a geometry/text/style oracle *lossless with respect to the declared IR* straight from the source file (the IR→render gap that G7 exploits is exactly what this oracle does *not* capture, by design), with no human or self-annotation bias. From 26 real decks we inject one single-shape defect per slide *in PPTX XML space* (overflow, overlap, alignment offset, font, margin) and render both the clean and the defective one-slide deck with the same real renderer (LibreOffice), so the paired pixel diff isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs / 3 families). Two results follow. First, the §4 "snap absorbs 45%" hazard is *template-specific*: real free-form decks absorb only 7% (render-fidelity 0.93), so the perturbation is faithfully present, a positive control on that claim. Second, *all three* attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are image-sufficient (balanced accuracy A 0.70–0.74, structure adds nothing); a margin bleed is a perception bottleneck the oracle rescues for the weaker VLMs (A 0.59 → B 0.70), the "missing structure, not missing capability" cell; and fine alignment stays near chance in *every* channel for *every* model (A = B = C ≈ 0.50; the internal-contrast re-run at a supra-threshold 44 px offset reproduces this, A 0.54/B 0.51/C 0.52, n=41). This is *not* a hard capability floor: on clean synthetic slides the same alignment defect recovers under targeted/relative elicitation once it clears acuity (Fig. 5); but on real cluttered layouts the misaligned element cannot be isolated among many genuine ones and specificity collapses, so for declared geometry the symbolic linter (which reads the coordinates) remains the reliable detector regardless (Fig. 11).

Open-world: structure recovered from pixels does not restore the linter. The hybrid's symbolic engine needs the declared IR a `.pptx` carries; a third-party deck exported to PDF/PNG ships none, so the natural question is

Table 3: Multi-reward / VLM-judge audit: paired preference accuracy $P(r_{\text{clean}} > r_{\text{def}})$ per class (chance = 0.5; G7 with 95% CI). Five published scorers in two audited categories each (general-multimodal: Skywork-VL + PickScore on distinct backbones; aesthetic: LAION + CLIP-IQA) plus one closed frontier VLM judge, so the G7 read is not a single-scorer anecdote. ✓/× marks reliable G7 detection (CI above / spanning 0.5). The VLM-judge row is a small closed-model probe (n=24); other n in the bottom row. G5 is the internal-chromatic re-operationalisation of Sec. 4 (one element hue-swapped vs. its sibling column), a *visible* defect most scorers catch, not the linter-only blind spot G7 is.

Reward scorer	category	G1	G2	G5	**G7**	S4	S6
DocReward-3B	document	0.93	1.00	0.72	× **0.48** [.38, .58]	1.00	1.00
Skywork-VL-7B	general-mm	0.94	0.72	0.57	✓ **0.79** [.69, .86]	0.92	1.00
PickScore-v1	general-mm	0.74	0.59	0.70	✓ **0.71** [.61, .80]	1.00	0.72
LAION-Aesth.	aesthetic	0.37	0.91	0.78	× **0.57** [.46, .66]	0.75	0.61
CLIP-IQA	aesthetic	0.44	0.50	0.65	✓ **0.83** [.74, .90]	0.50	0.72
Qwen3.7-Max judge	frontier VLM	0.13	0.92	–	✓ **1.00** [.86, 1]	1.00	–
n (paired; non-VLM rows)		54	54	80	90	36	18
n (Qwen3.7-Max judge row)		24	24	–	24	24	–

Table 4: Page-semantic inspection, balanced accuracy [95% Wilson] on paired clean (best modality). The finetuned 8B examiner beats both zero-shot baselines, including the 30B, *in-distribution*. Deck-scope S2/S5 are a negative result (text). n = paired positives (2-AFC rows = forced-choice pairs); the S6 2-AFC rests on only 12 pairs and is descriptive only. "Best modality" denotes the diagnostic-prescribed modality (oracle channel B for geometry, image-only for semantics), pre-determined by the attribution logic, not selected post-hoc.

class	**ft-8B**	zs-8B	zs-30B	n
S1 title/body	**1.000** [.95,1]	0.778 [.69,.85]	0.933 [.87,.95]	36
S4 density	**1.000** [.97,1]	0.529 [.50,.58]	0.774 [.69,.84]	60
G1 overflow (2-AFC)	**1.000** [.97,1]	0.975 [.93,.99]	0.700 [.61,.77]	120
S6 image/text (2-AFC)	**1.000** [.76,1]	1.000 [.76,1]	0.500 [.25,.75]	12
S2 narrative (deck)	0.500† [.45,.53]	0.646 [.53,.72]	0.549 [.42,.66]	36
S5 missing-logic (deck)	0.500 [.47,.53]	0.500 [.47,.53]	0.567 [.45,.68]	60

† degenerate (always-flag): the SFT mix had no clean-*deck* negatives. See text.

whether *recovering* element structure from the rendered pixels can re-arm the linter. We test it: following the DocLayNet/PubLayNet line of document-layout detection datasets and benchmarks [20, 21], a transformers-native detector (PP-DocLayoutV2) returns region boxes on each render, class-agnostic NMS removes the detector's cross-class duplicate boxes (which would otherwise manufacture phantom overlaps), and the boxes, normalised into the IR coordinate frame, feed the *same* geometry linter at its shipped operating point. On the real-layout set (GT IR *and* render available, so the full ladder and a recovery-fidelity IoU coexist) the answer is a pre-registered negative (Fig. 13): recovered structure leaves the linter at chance on G1/G3/G4 (overflow needs text capacity, alignment needs declared positions, font needs point sizes, none surviving a pixel→box projection), and rescues only overlap partially (G2 balanced accuracy 0.54 vs. native-IR 0.67 vs. VLM-only 0.87). The detector *over-segments* (≈ 2× more boxes than the GT IR, recovery recall at IoU≥0.5 only 0.30–0.39), so genuine element collisions are split apart and overlap recall falls 0.78 → 0.29. The same picture holds on image-only SlideAudit (no IR at all): the recovered linter clears chance only on G2 (0.55) while the C3-elicited VLM reaches 0.94. Two points sharpen the deployment scope. First, on real third-party decks even *native* IR is weakened: the decks carry no expected-position/font bookkeeping, so the linter's G3/G4 rules are also at chance, and its margin/overlap rules *over-fire* on tight real layouts (specificity 0.11/0.56), eroding the near-zero-false-positive advantage it enjoys on clean synthetic IR. Second, as a consequence the VLM-only engine *dominates every coarse class* on real decks. The linter's edge is an in-distribution, clean-native-IR phenomenon; open-world, recovered or not, the deployable critic is VLM-bound. This bounds the contribution without touching the diagnostic thesis: the full symbolic–neural critic is for IR-owning generation agents, and for bare third-party pixels the critic is the C3-elicited VLM.

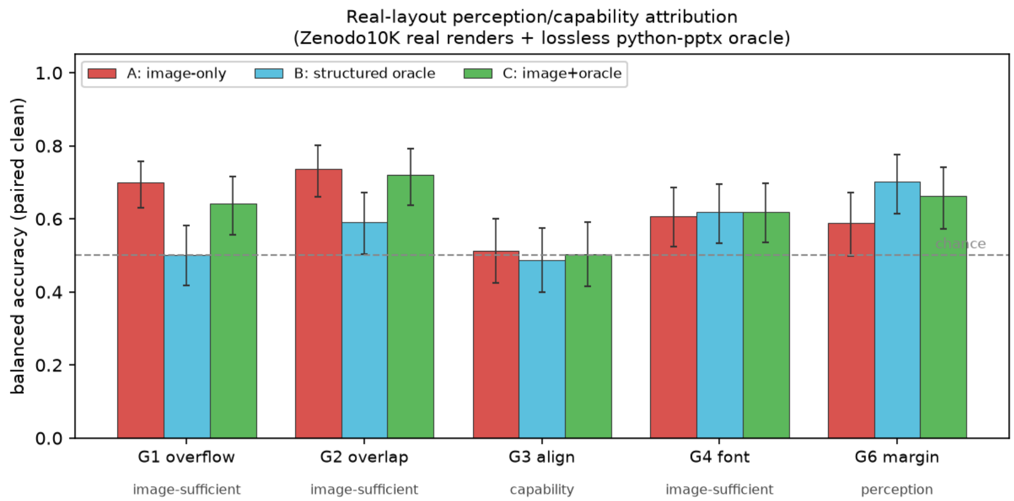

Figure 11: The perception/capability split reproduces on real layouts (ZENODO10K real renders + lossless `python-pptx` oracle; mean over 3 VLMs / 3 families, balanced accuracy on paired clean; error bars = 95% Wilson on counts pooled across models, n=76–90 paired per class). Coarse defects (G1 overflow, G2 overlap) are image-sufficient; G6 margin is a *perception* bottleneck the structured oracle rescues (B>A); G3 alignment stays near chance in every channel on these real cluttered decks (A=B=C≈0.5), not a hard capability floor (it recovers on clean slides above threshold, Fig. 5) but unreliable enough on real layouts that it must route to the linter. Intervals on G3/G4/G6 include chance, consistent with their small per-class n (see released reports).

8 Limitations

We now report a human spot-check on the rebuilt corpus, but not an inter-annotator or deployment-time human baseline. The paper's contribution is the *diagnosis* (which bottleneck a defect hits) and the routing that follows from it; matching or exceeding human inspectors is a different claim we do not make. On the rebuilt v2 sample, every sampled defect is judged visible and every clean twin clean (73/73; the auditable structure-carrying subset is 55/55 source-faithful), so the paper's use of G3/G5 rests on corrected human-verified samples rather than the stale legacy framing.

Three scope bounds qualify the evidence. The real-layout attribution (§7) uses *injected* defects (real geometry, renderer, and oracle, but not naturally-occurring), so the image-only, naturally-defective SLIDEAUDIT set remains our complementary real probe. The reward audit covers five published scorers and one closed judge; its document-reward miss is instance-level, and a second document/design reward (PosterReward) and an InternLM-family reward (PloRA) were not stood up, so the categorical read rests on the two general-multimodal backbones and the same-backbone CLIP-IQA/LAION dissociation. Last, the examiner→generation *downstream* effect is near-null: examiner quality correlates +0.66 with self-refine gain under a strong generator but at sub-1% magnitude, and collapses to 0.00 under an unfloored weak generator (1.43× the headroom); a controlled A/B locates this as an axis mismatch, since the weak generator's headroom is deck *coverage*, which a layout critic does not move (quality change Δq=+0.00 vs. a coverage critique's +0.32), so the examiner's evidence is its intrinsic and case-study performance, not this loop.

Two caveats temper the central claims, both controlled in the body. First, the forced-choice and atomic-binary elicitations supply a contrastive reference and a named target the pointwise format withholds; the C0+ and guess-floor controls (§5.1, Fig. 8) therefore scope strong format suppression to G7 and identify G1/S6 as reference-assisted. Second, the 0.86 mean balanced accuracy and strict 6/9 coverage are *native-structure* results: on image-only real decks the hybrid degrades to VLM-only, and recovering structure from pixels does not close the gap (§7, Fig. 13). They are therefore an upper bound for IR-owning generation agents; on bare third-party pixels the deployable critic is the C3-elicited VLM.

9 Conclusion

A slide-defect critic should not be one model. Telling apart sub-perceptual, format-suppressed, and reference-assisted failures turns "VLMs are bad at layout" into a routing rule: declared geometry to a symbolic linter, targeted single-slide elicitation for the format-suppressed classes, and paired reference use where a clean twin is what resolves the defect. A linter plus one re-elicited VLM reaches mean balanced accuracy 0.86 against the conventional VLM-C0's 0.59, with 6/9

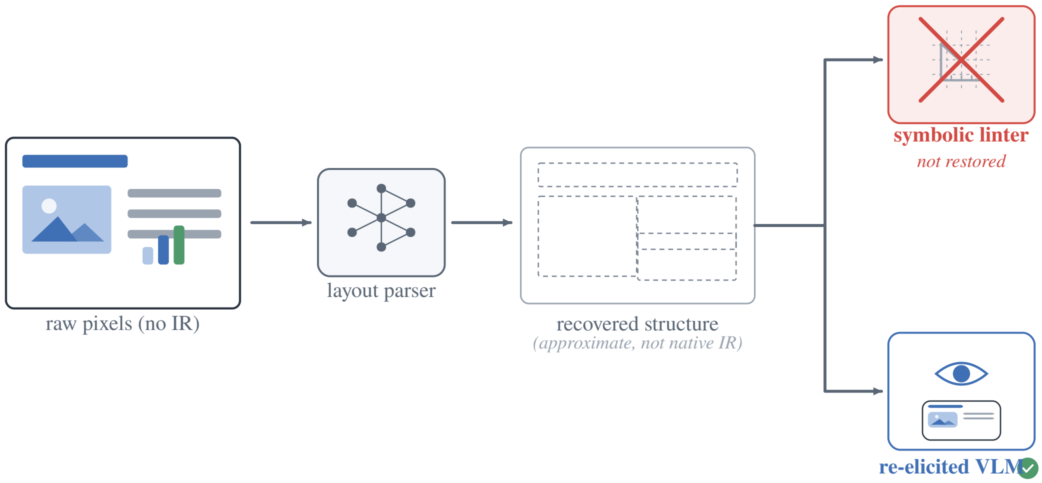

Figure 12: Open world: recovered structure does not restore the linter. For bare third-party pixels with no IR, a layout parser recovers approximate structure, but it is not the native IR the symbolic linter needs; only the re-elicited VLM remains a deployable critic (quantified in Fig. 13).

image-level classes strictly covered. A constructed render-overflow class makes the linter's blind spot falsifiable; across the scorers tested, its detection tracks a rendered-quality read-out rather than a domain label. The contribution is the lossless-oracle diagnosis from which the routed critic follows, not a claim that one engine solves every class.

A SlideAudit crosswalk

Reproducibility

All analysis scripts, the defect injector, the two linters, the elicitation harness (including the C0+ coverage-vs-format control, `part3_elicit.py -condition C0plus`), the hybrid router, the multi-reward audit adapters, the real-layout (Zenodo10K) IR/inject/render pipeline, and the per-Part reports are released. A reproducibility bundle with paired manifests and per-item rollout CSVs is available under `release/part3/` (generated by `scripts/part3_reproducibility_release.py`). Experiments ran on 4×RTX 3080 (20 GB) with vLLM-served open VLMs (Qwen3.5/3.6, Gemma4, InternVL3.5, Ovis2.5), a QLoRA-finetuned Qwen3-VL-8B, five published reward scorers, and one DashScope Qwen3.7-Max VLM-judge probe (DocReward-3B; Skywork-VL-Reward-7B and PickScore; a LAION aesthetic predictor and CLIP-IQA) behind a uniform adapter; serving recipes (tensor-parallel, AWQ, disabled thinking, fp8-KV caveats on Ampere) are in the repository. Weights, renders, raw rollouts, and upstream licensed items are not redistributed; these omissions are licensing-constrained, not convenience-constrained. Every balanced-accuracy cell in the tables and figures carries its 95% Wilson interval and per-cell n; the full family of paired tests is reported with Holm (FWER) and Benjamini–Hochberg (FDR) multiplicity correction over all 61 reported significance tests (released multiplicity report), and headline claims are stated as surviving or downgraded under that correction.

References

[1] Liu et al. On the Perception Bottleneck of VLMs for Chart Understanding. arXiv:2503.18435.

[2] Fu, Bonnen, Guillory, Darrell. Hidden in Plain Sight: VLMs Overlook Their Visual Representations. arXiv:2506.08008.

[3] Tong et al. Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. CVPR 2024; arXiv:2401.06209.

[4] Wang et al. Your Reasoning Benchmark May Not Test Reasoning: Revealing Perception Bottleneck in Abstract Reasoning Benchmarks. arXiv:2512.21329.

[5] Heo, Hwang, Jung, Jung. LED Benchmark: Diagnosing Structural Layout Errors for Document Layout Analysis. arXiv:2507.23295.

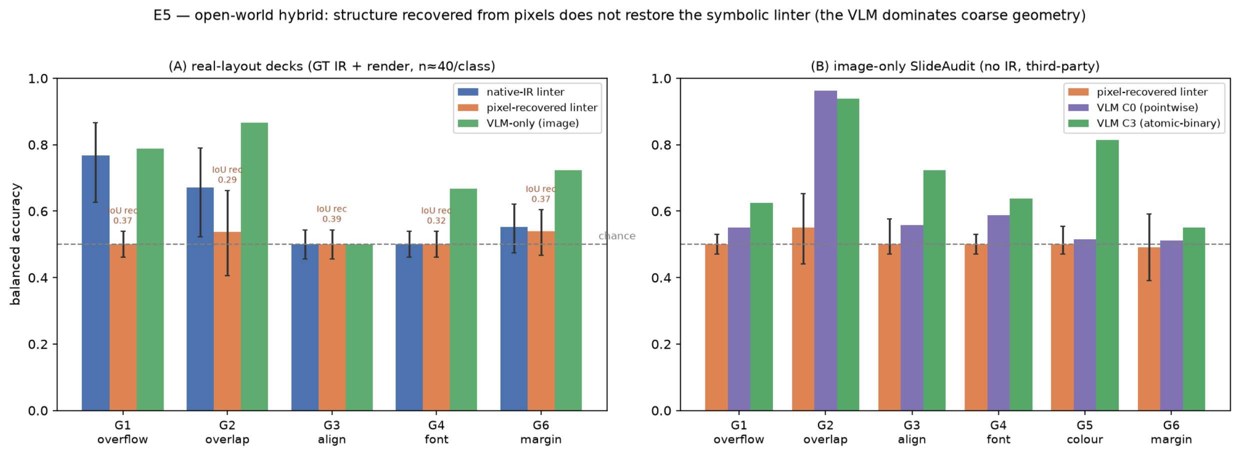

Figure 13: Open-world hybrid: structure recovered from pixels does not restore the symbolic linter. A document-layout detector (PP-DocLayoutV2) recovers element boxes from the render; the *same* geometry linter runs on them. **(A)** real-layout decks (GT IR + render, $n\approx40$/class): native-IR linter vs. pixel-recovered linter vs. VLM-only floor, Wilson CIs; recovered structure rescues only overlap (G2) and is silent on fine geometry, while the VLM dominates every class. **(B)** image-only SlideAudit (no IR, third-party): the recovered linter clears chance only on G2; the C3-elicited VLM is far stronger. The linter's near-zero-FP advantage requires clean native IR.

[6] Kumar et al. VLM Judges Can Rank but Cannot Score: Task-Dependent Uncertainty in Multimodal Evaluation. arXiv:2604.25235.

[7] Liusie, Manakul, Gales. LLM Comparative Assessment: Zero-shot NLG Evaluation through Pairwise Comparisons using Large Language Models. EACL 2024; arXiv:2307.07889.

[8] Tripathi, Wadhwa, Durrett, Niekum. Pairwise or Pointwise? Evaluating Feedback Protocols for Bias in LLM-Based Evaluation. arXiv:2504.14716.

[9] Pan et al. AeSlides: Incentivizing Aesthetic Layout in LLM-Based Slide Generation via Verifiable Rewards. arXiv:2604.22840.

[10] Liu et al. Presenting a Paper is an Art: Self-Improvement Aesthetic Agents for Academic Presentations (EvoPresent/PresAesth). arXiv:2510.05571.

[11] Ge et al. AutoPresent: Designing Structured Visuals from Scratch. CVPR 2025; arXiv:2501.00912.

[12] Kang, Bao, Goswami. VLM-SlideEval: Evaluating VLMs on Structured Comprehension and Perturbation Sensitivity in PPT. arXiv:2510.22045.

[13] Zhang, Chen, Zhong, Wobbrock. SlideAudit: A Dataset and Taxonomy for Automated Evaluation of Presentation Slides. arXiv:2508.03630.

[14] Liu et al. DocReward: A Document Reward Model for Structuring and Stylizing. arXiv:2510.11391.

[15] Wang et al. Skywork-VL Reward: An Effective Reward Model for Multimodal Understanding and Reasoning. arXiv:2505.07263; weights `Skywork/Skywork-VL-Reward-7B`.

[16] LAION-AI. Aesthetic Predictor: a linear estimator on CLIP for image aesthetic quality (LAION-Aesthetics). Software, 2022. https://github.com/LAION-AI/aesthetic-predictor.

[17] Kirstain et al. Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation. NeurIPS 2023; arXiv:2305.01569; weights `yuvalkirstain/PickScore_v1`.

[18] Wang, Chan, Loy. Exploring CLIP for Assessing the Look and Feel of Images. AAAI 2023; arXiv:2207.12396.

[19] Zheng et al. PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides (introduces the Zenodo10K corpus of real `.pptx`). arXiv:2501.03936; dataset `Forceless/Zenodo10K`.

[20] Pfitzmann et al. DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. KDD 2022; arXiv:2206.01062.

[21] Zhong, Tang, Jimeno Yepes. PubLayNet: largest dataset ever for document layout analysis. ICDAR 2019; arXiv:1908.07836.

Table 5: Crosswalk between our G/S classes and the 19-dimensional SlideAudit taxonomy. Empty entries are intentional off-taxonomy semantic or deck-level defects; G7 refines, rather than duplicates, SlideAudit's content-overflow/cut-off category by adding the declared-legal-box criterion that makes the defect linter-blind.

Our class	SlideAudit class	Relation
G1 text overflow	Content Overflow/Cut-off	equivalent
G2 element overlap	Occluded Content	equivalent
G3 alignment offset	Content Alignment Issues	equivalent
G4 font-size inconsistency	Improper Font Sizing	equivalent
G5 brand-color violation	Excessive or Inconsistent Color Usage; Inappropriate or Mismatched Color Combinations	equivalent
G6 margin violation	Unbalanced Space Distribution	near-equivalent; safe-margin bleed is a special case
G7 render-containment overflow	Content Overflow/Cut-off; related to Improper Image Sizing	our extension; same visual symptom plus declared-legal-box criterion
S1 title/body mismatch	–	off-taxonomy content semantics
S2 narrative-order break	–	off-taxonomy deck-level ordering
S3 terminology inconsistency	–	off-taxonomy cross-slide terminology
S4 density violation	Excessive Text Volume	equivalent
S5 missing logic section	–	off-taxonomy deck-level completeness
S6 image/text contradiction	Irrelevant Visual Content (nearest)	nearest only; contradiction rather than relevance